

| Question |     | Answer                    | Mark | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                                                                          |
|----------|-----|---------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15       | (a) | 488 to 507                | M1   | for method to find area of one strip using trapezia,<br>eg $\frac{1}{2} \times 5 \times 22$ (= 55) or $\frac{1}{2} \times 5 \times (22 + 28)$ (= 125)<br>or $\frac{1}{2} \times 5 \times (28 + 32)$ (= 150) or $\frac{1}{2} \times 5 \times (32 + 35)$ (= 167.5)<br><b>OR</b><br>for a method to find an estimate for the area using rectangles<br>eg $5 \times 22$ or $5 \times 28$ or $5 \times 32$ or $5 \times 35$ | May use area of triangle + area of rectangle for the second, third and fourth strips – lengths must be correct.<br><br>May use triangle for first strip,<br>$\frac{1}{2} \times 5 \times 22$ |
|          |     |                           | M1   | for complete and correct method to find the area using four strips,<br>eg $\frac{1}{2} \times 5 \times 22 + \frac{1}{2} \times 5 \times (22 + 28) + \frac{1}{2} \times 5 \times (28 + 32)$<br>$+ \frac{1}{2} \times 5 \times (32 + 35)$<br>or $5 \times 22 + 5 \times 28 + 5 \times 32 + 5 \times 35$                                                                                                                  |                                                                                                                                                                                              |
|          |     |                           | A1   | for answer in the range 488 to 507<br><br>(SC B1 for using area under the curve)                                                                                                                                                                                                                                                                                                                                       | May use triangle for first strip,<br>$\frac{1}{2} \times 5 \times 22$                                                                                                                        |
|          | (b) | Underestimate (supported) | C1   | (dep M1) for underestimate since parts not included below the graph<br><b>OR</b> ft their method                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                              |